ConceptGate: Few-Shot Concept Reading and Steering in Frozen Language Models

Nishan Pantha
Independent Researcher
nishanpantha@gmail.com

Abstract

Deployed language models must detect and control concepts such as a jailbreak attempt, a topic, or a tone. Often they have few examples and cannot retrain. One residual-stream direction can detect a concept, and the same direction, added back, can steer generation. We test how far this goes with about ten examples per class and closed-form fitting. ConceptGate reads a concept at several layers, fuses the layers with a learned depth filter, gates on a calibrated likelihood ratio, and steers with a direction fit from the same examples. Against fair baselines on five models up to 2.6B parameters, most of the design fails. A logistic probe on the same layers matches the detector. Depth fusion helps only on synthetic data. Mixture densities collapse to one component. A gate on concept presence selects nothing, because it fires on 97% of formatted attacks. One early result favoured the gate, but it came from a missing chat template. With the template, the steering effect changed sign and the gate's advantage disappeared. Three results survive. First, the few-shot direction is a refusal lever in both directions (Qwen2.5-0.5B refusal: 64% unsteered, 35% steered away, 79% steered toward). Second, the gate's false-positive rate changes from 18% to 96% with the register of benign traffic. Third, on one of two models, the unsteered activations predict a prompt's steerability (Spearman $\rho = 0.58$, permutation $z = 6.7$). Steerability is how far a fixed write moves the prompt's sampled refusal rate. The concept gate reads it at $|0.30|$, with the wrong sign. As a trigger, predicted steerability beats random selection at every tested coverage. It does not replicate on gemma-2-2b, where more samples made the correlation fall. We also show how an unvalidated outcome measure gave a false null, and why split-half reliability is not comparable across sample sizes.

1 Introduction

A frozen language model encodes many nameable properties of its input along approximately linear directions in its residual stream [1, 21, 28]. A direction can be read: this gives a probe. A direction can also be written back: this gives a steering vector [20, 25]. This paper asks what we can build from these two facts under tight constraints:

- no fine-tuning of the host model;
- about ten labelled examples per concept;
- closed-form fitting;
- kilobytes of storage per concept.

Our system, ConceptGate, attaches to a frozen model. It does two things with one set of examples. It *detects* a concept from intermediate activations and returns a calibrated fire, abstain, or pass decision. It *steers* generation toward or away from the concept: it adds a related direction

into the residual stream. The first application was guardrails, but the mechanism is not specific to safety. The concept is an input, given as data.

The design had two hypotheses:

1. **Read across depth.** A read over several layers is better than a read at the best single layer. If the layers carry independent evidence, a matched filter over depth combines them, and their discriminabilities add in quadrature.
2. **Couple the read and the write.** If the write fires only where the read detects the concept, steering is more effective and causes less damage than steering on every prompt.

Both hypotheses failed in the form we stated them. This paper reports each component against a fair baseline. It shows which tests failed. It shows one strong result that came from an error in prompt formatting. It also shows one positive result that came when we asked a different question of the same system. We give the negative results the same weight as the positive results, because they contain most of the useful information.

Summary of findings.

1. **The read side gives no advantage over a probe.** A depth-matched logistic probe equals ConceptGate at every tap (section 5.4). Depth fusion gets its predicted gain on synthetic data that satisfies its assumption (test error 16.1%→9.4%). It adds nothing on real models (section 5.1). The mixture density stays at one component (section 5.2). A training-free concept bank is 30–38× cheaper to build than per-concept LoRA. A linear-probe bank has the same advantage (section 5.5).
2. **The write is a real lever, but not a new one.** Eight hand-written examples move generated refusal across a 43-point range, in either direction. On a first-token measure, the effect is about three times that of a matched-norm random direction (section 6). This is the refusal direction of Arditi et al. [2], recovered from few examples. It is not a new mechanism.
3. **A gate on concept presence adds nothing. A gate on predicted effect helps, on one model.** On formatted attacks, the concept is almost always present, so the concept gate cannot select. A ridge read fit to the measured per-prompt effect of the write can select. On Qwen2.5-0.5B, it beats a size-matched random subset at every coverage. It does not replicate on gemma-2-2b (section 7).
4. **Three cautions on method.**
   (a) Prompt formatting alone reversed the sign of the main gate-steering effects. It also removed the apparent advantage of the gate (section 6.3).
   (b) An outcome measure with unknown reliability can make a false null (section 7.3).
   (c) Split-half reliability is not comparable across sample sizes (section 7.7).

No single mechanism here is new. Probes, difference-of-means directions, activation addition, Gaussian density gates, and steering vectors conditioned on an activation read [6, 15] all exist. Our contribution is the measurement. We map where a few-shot concept adapter works and where it does not. We also give one possibly new result on per-prompt steerability.

2 Related work

Probes and representation engineering. Linear probes read what a layer encodes linearly [1]. Representation engineering [28] states that high-level concepts lie along directions that a user can read and push. Its reading vectors come from PCA on paired activation differences. Our class-mean difference is the construction of contrastive activation addition [20]. Most such work uses one layer, selected by a sweep. Our only change on the read side was to fuse several layers. It gave no gain (section 5.1).

Activation steering and conditional steering. ActAdd [25] adds a contrastive direction to the residual stream. We use the form of Contrastive Activation Addition [20]. Circuit Breakers [29] trains the model to reroute harmful representations. We steer a frozen model, so we accept less power for less cost. CAST [15] conditions a steering vector on an activation read. It applies or holds a refusal vector based on a condition read from the prompt's hidden states. Thus the model refuses harmful prompts and answers harmless prompts. DSAS [6] also separates when to steer from how to steer. It scales the strength of an existing steering method per input and per layer. Thus the setting of section 6 is not new. Our contribution there is the controls. One direction mediates refusal [2]. The model encodes harmfulness separately from refusal [27]. Steering along random or unrelated directions can decrease refusal [13, 18]. CAST's condition rules classify mathematics problems as malicious and refuse them [24]. Activation-trajectory detectors for multi-turn attacks do not generalize equally across data sources [14]. Our results agree with each of these observations. We get them from about ten examples.

Density-based detection. The Mahalanobis out-of-distribution detector [16] scores with class-conditional Gaussians. Our gate is a two-class case of it. Each class can optionally be a mixture, with the number of components selected by BIC.

External classifiers and cost. The current standard for detection is a text classifier, such as Llama Guard [10], or a fine-tuned multi-category guardrail on an encoder such as GLiNER [26]. These are strong. But they are a second model, and they can only read. To run only the layers below the deepest tap is the detection equivalent of early exit [23]. Every latent probe gets this saving. For the task of adding concepts to a model, the comparison is parameter-efficient fine-tuning [8, 9].

Predicting steering outcomes. Fan et al. [5] train a classifier on ASTEER, their testbed of labelled steered generations. The classifier predicts if an intervention will under-steer, succeed, or over-steer. Its features compare hidden states before and after steering over the first decoding steps. Billa [4] uses a logit-lens diagnostic to predict, per concept and per layer, if a difference-of-means steering vector will be effective. Ong et al. [19] use unsteered representations to predict the side effects of a steering vector on other behaviours. Our first-token refusal-minus-compliance statistic is a basket-sum variant of the refusal–affirmation logit gap of Li and Liu [17]. The Cylindrical Representation Hypothesis [7] gives a geometric explanation of why steering sensitivity changes from sample to sample. It does not predict the sensitivity from the unsteered prompt. We found no prior work that predicts, from the unsteered prompt alone, how far a fixed write will move the behaviour on that prompt.

Adversarial limits. Obfuscated-activation attacks decrease the recall of latent-space probes to zero, and the target behaviour continues [3]. This limit applies to all results in this paper (section 9).

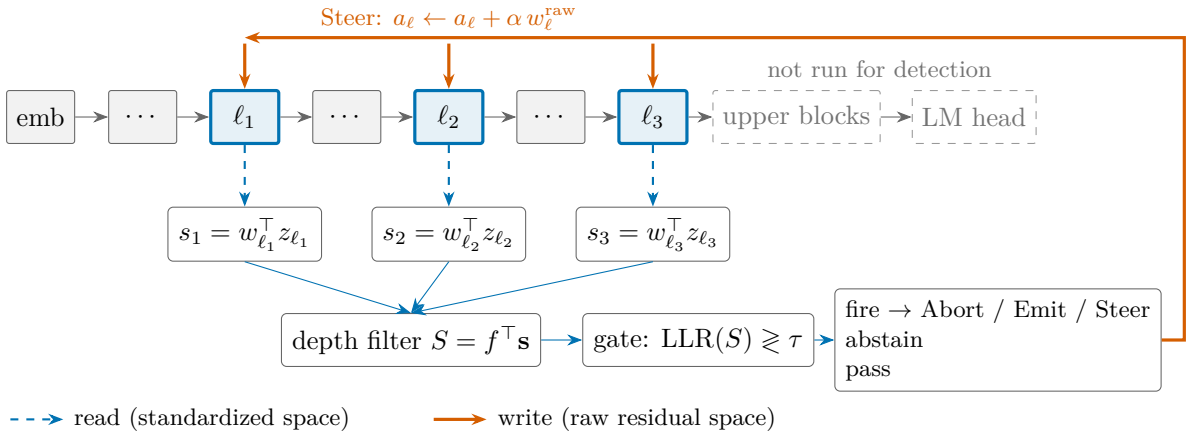


Figure 1: The ConceptGate pipeline. The frozen model runs up to the deepest tap. We standardize each tapped activation and project it onto the concept direction of that layer. This gives one score per layer (the *spectrogram* $\mathbf{s}$). A depth filter combines the scores. A calibrated likelihood-ratio gate returns fire, abstain, or pass. On a fire, an action can stop decoding, start a fixed continuation, or add a raw-space direction back into the stream. The same examples fit this direction. The read and write directions are related but not identical (per-tap cosine 0.45–0.83, section 6.2).

3 Method

Figure 1 shows the full path. All read-side fitting is closed form.

3.1 Setup and standardization

M is a frozen causal language model with residual width d. We tap block outputs at layers $\mathcal{L} = \{\ell_1, \ldots, \ell_m\}$. Each prompt is represented by its *last-token* activation $a_\ell \in \mathbb{R}^d$. This token has attended to the full prompt. A fit on every token adds shared boilerplate, which hides the signal. The positive and negative sets $\mathcal{A}^\pm$ each contain about ten prompts. Residual streams have a few very large outlier dimensions. Thus, for detection, we standardize per layer and per dimension with pooled statistics, $z = (a - \mu_0) \oslash (\sigma_0 + \epsilon)$.

3.2 Directions and the spectrogram

The per-layer detection direction is the unit difference of the standardized class means:

$$w_\ell = \frac{\bar{z}_\ell^+ - \bar{z}_\ell^-}{\|\bar{z}_\ell^+ - \bar{z}_\ell^-\|}. \tag{1}$$

In *logistic* mode, it is a per-layer logistic-regression weight. Diff-of-means equals the LDA direction $\Sigma^{-1}(\mu^+ - \mu^-)$ only if the standardized dimensions are uncorrelated. Standardization makes the marginal variances equal, but it does not remove correlations. The logistic mode closes this gap. The projection at each layer, $s_\ell = w_\ell^\top z_\ell$, gives the profile of the concept across depth. We call this profile the spectrogram $\mathbf{s} \in \mathbb{R}^m$.

3.3 Depth filter and the quadrature argument

A filter $f \in \mathbb{R}^m$ reduces the spectrogram to one score, $S = f^\top \mathbf{s}$. We use three filters:

- `best`: a one-hot on the most discriminative layer. This is the single-layer baseline.

- `diag`: each layer has a weight equal to its own signal-to-noise ratio.
- `fisher`: $f \propto \Sigma_{\mathbf{s}}^{-1}(\bar{\mathbf{s}}^{+} - \bar{\mathbf{s}}^{-})$, with a ridge-regularized pooled covariance.

`best` is a nested special case. Thus a comparison of the filters isolates the effect of depth.

The motivation is a matched-filter argument. Let $s_\ell = a_\ell y + n_\ell$, with $y \in \{\pm 1\}$ and independent noise $n_\ell \sim \mathcal{N}(0, \sigma_\ell^2)$. Then the filter $f_\ell \propto a_\ell / \sigma_\ell^2$ gives the maximum discriminability after the combination, and

$$d'_{\text{comb}} = \Big(\sum_{\ell} {d'_\ell}^2 \Big)^{1/2} \;\geq\; \max_{\ell} d'_\ell, \tag{2}$$

with equal-prior error $\Phi(-d'/2)$. The critical assumption is independence across layers. Adjacent residual-stream layers are not independent. This is why the argument fails in practice (section 5.1).

3.4 Calibrated gate and mixtures

We fit class-conditional densities to the score. The gate fires when LLR $= \log p(\cdot \mid +) - \log p(\cdot \mid -) > \tau$. We set the threshold on the benign tail. Two rules are available: a false-positive quantile, or the point z standard deviations above the benign mean ($z = 3$ is approximately a 0.1% tail). We map the LLR to $P_{\text{present}} = \sigma((\text{LLR} - \tau)/s)$, where s is the benign spread near τ. The gate abstains when $|P_{\text{present}} - \frac{1}{2}|$ is less than a margin. As an option, each class is a Gaussian mixture on the joint spectrogram, with $J \in \{1, 2, 3\}$ components selected by BIC. With $J = 1$ and a shared covariance, the LLR is affine in $\mathbf{s}$. It then reduces exactly to the `fisher` filter.

A bank of K concepts fires if one or more concepts fire. It assigns the fire to the concept with the largest LLR. Per-concept thresholds do not compose. An OR over $K = 14$ concepts at $z = 3$ gives a bank-level false-positive rate of up to approximately 1.4%. Thus the operating point must be set for the full bank.

3.5 Steering

The steering hook sees raw activations. Thus the write uses a separate raw-space direction per tap, $w_\ell^{\text{raw}} = (\bar{a}_\ell^{+} - \bar{a}_\ell^{-})/\|\bar{a}_\ell^{+} - \bar{a}_\ell^{-}\|$. During generation, the hook applies $a_\ell \leftarrow a_\ell + \alpha\, w_\ell^{\text{raw}}$. A positive α steers toward the concept. A negative α steers away from it. The residual norm of GPT-2 is approximately five times that of Qwen2.5-0.5B (96 and 18 in our runs). Thus we set α as a fraction of the measured residual norm. In development runs, 3–10% kept the text coherent. Above approximately 20–25%, the text became repetitive.

3.6 Actions, triggers, and the truncated forward

Detection and steering use one interface. An action (`Abort`, `Emit`, `Steer`) runs when its trigger is true. The triggers are `FIRE`, `FIRE_OR_UNSURE`, and `ALWAYS`. We added two triggers after the work in section 7. `Predicted(name, τ)` fires on the prediction of an outcome head, not on concept membership. `Both(a, b)` requires both conditions. Detection needs only blocks $0..\max \mathcal{L}$. On GPT-2, the truncated forward is bit-identical at the taps and 1.75× faster than a full forward. A weight-truncated load mode never loads the upper blocks. Generation still needs the full model. Section A lists the library calls.

3.7 Cost

Table 1 lists what one concept stores. For GPT-2 with five taps, the total is approximately 1.5×10^4 numbers. For the three-tap configurations below, it is 11–28 thousand. A fit is a few means and one

$m \times m$ solve. It takes milliseconds on a CPU.

Table 1: Storage per concept (m taps, residual width d).

Item	Numbers
read and write direction sets	$2md$
standardization statistics	$2md$
depth filter	m
gate	a few scalars

4 Experimental setup

Models. GPT-2 (12 blocks, float32), Qwen2.5-0.5B-Instruct (24 blocks), and gemma-2-2b-it (26 blocks). One sweep also uses SmolLM2-360M-Instruct and SmolLM2-1.7B-Instruct. Instruct models load in bfloat16 unless stated. The first GPT-2 and Qwen detection results ran on a CPU. All other runs used an Apple M4 GPU (MPS).

Data. Table 2 lists the data for each experiment.

Table 2: Data per experiment.

Experiment	Data
Jailbreak detection	`jackhhao/jailbreak-classification` [11]. Fit on the training split. Scored on the official test split (262 prompts: 139 jailbreak, 123 benign).
Multi-concept bank	The fourteen harm categories of BeaverTails [12]. Positives are prompts whose responses have that annotation. Negatives come from a shared benign pool.
Topical steering	Hand-written few-shot sets for *food*, *nature*, and *technology*.
Steering gate (sections 6 and 7)	Hand-written short override framings; held-out jailbreak templates with a harmful request appended; bare harmful requests; real benign prompts.

Prompt formatting. Detection benchmarks read raw prompts. From section 6.3 onward, all steering and gate experiments format prompts with the chat template of the model, for reading and for generation. Section 6.3 explains why this is important.

Statistics. Unless stated, $\pm$ is the standard deviation over three few-shot resamples. Correlations are Spearman. Permutation nulls run the identical pipeline again on shuffled targets.

5 The read side: a sequence of negative results

5.1 Depth fusion works only where its assumption holds

The synthetic data has three independent layers with known $d' = [1.6, 2.0, 0.6]$. Theory predicts a fused $d' = 2.63$ and a test error of $\Phi(-1.315) = 9.4\%$. The best single layer gives $\Phi(-1.0) = 15.9\%$. The learned filter recovers $d' = [1.62, 2.04, 0.64]$. It decreases the test error from 16.1% to 9.4%. This result shows only that the code implements the algebra correctly. We generated the data under the assumption of the filter, so the result is almost a tautology.

Table 3: Components selected by BIC (positive / negative class, median of three seeds) as the fit size N per class increases. Three taps at 33/50/67% depth. The last row is the largest held-out AUC gain of the BIC-selected gate over the same gate with $J = 1$, across both models and both modes.

N per class	12	32	64	128	256
Qwen2.5-0.5B, diff-of-means	1/1	1/1	1/1	1/**2**	**2**/**2**
Qwen2.5-0.5B, logistic	1/1	1/1	1/1	**2**/**2**	**2**/**2**
gemma-2-2b, diff-of-means	1/1	1/1	1/1	1/1	**2**/**2**
gemma-2-2b, logistic	1/1	1/1	**2**/1	**2**/**2**	**2**/**2**
best held-out AUC gain	0.000	0.000	+0.001	+0.006	+0.007

On real models, the gain disappears (fig. 3, table 4). Three-tap and five-tap fusions are equal to the best single tap. On Qwen2.5-0.5B, a logistic probe on the concatenated taps is slightly better than the ConceptGate fusion (0.978 and 0.973). The tapped layers are too correlated for quadrature to apply.

5.2 Mixture densities collapse

Mixtures can be important. We made a test case with benign clusters on the two sides of the harmful cluster. No linear threshold works on it (38.8% error, AUC 0.60). A two-component mixture gets 7.1% error (AUC 0.98; the Bayes floor is 5.8%). We made this case directly in score space. Its class means are the same, so our direction-finding cannot make it from real data.

On real GPT-2 activations with 12 prompts per class, BIC selects $J = 1$ for both classes. On held-out prompts, the mixture gate and the single-Gaussian gate agree at a rank correlation of 0.986. Sample size is not the only cause (table 3). On the jailbreak benchmark, in diff-of-means mode, BIC selects one component per class up to 64 examples per class on both models. This is far past the point where a three-dimensional covariance is estimable. When BIC selects a second component, the held-out AUC gain is 0.007 or less.

5.3 Matched negatives hurt, and a predicted robustness effect is absent

CAA [20] suggests that benign negatives in the same surface register as the jailbreaks make the direction more precise. We expected this result. On the development set (15 negatives of each type), matched negatives gave AUC 0.42. Broad negatives gave 0.78. At this sample size, we read 0.42 as a failure to separate, not as a stable inversion. Broad negatives let the direction align with the large semantic gap between an instruction to the model and an ordinary query. Matched negatives remove this gap.

We also expected that paraphrase would break shallow taps but not deep taps. That result would give a reason to tap deep. But rewording the positives has almost no effect on AUC at any layer. Against broad negatives, detection uses the same semantic gap, and rewording does not close it.

5.4 Detection gives no advantage over a probe

At each single tap, the ConceptGate logistic direction and a depth-matched logistic probe give the same number. The reason is that a single-tap logistic direction is logistic regression on that tap. Thus the fair single-concept comparison is a tie. The only gap is to a probe on the final layer, which runs the full model (table 4). At three taps, diff-of-means is 2–5 AUC points below logistic.

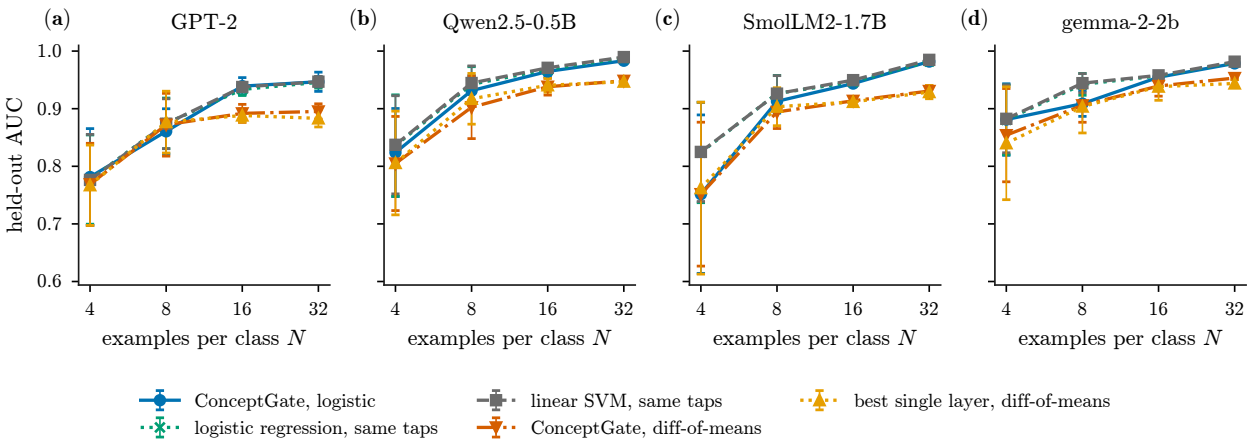


Figure 2: Sample efficiency on jailbreak detection: held-out AUC as the number of examples per class increases. Every method reads the same three tapped activations (mean ± s.d. over three resamples). Logistic regression on the taps is within 0.006 of ConceptGate-logistic at $N \geq 16$. At $N \leq 8$, where the resample spread is large, it is better by up to 0.07. Diff-of-means fusion is within ±0.015 of the best single layer at all N. This run is separate from table 4 and has its own resamples. Thus the values at $N = 32$ differ from that table by up to 0.01.

Thus every comparison with a trained classifier uses logistic mode. Figure 2 shows the same pattern on four models, at every sample size from four to 32 examples per class. Logistic regression on the same taps equals ConceptGate-logistic from 16 examples per class. Below 16, it is equal or better.

The truncated forward is a real saving over the full-model probe. On Qwen2.5-0.5B, one tap at 46% depth gets 0.970 and loads 61% of the weights. Every latent probe gets this saving. Thus it shows how early the model forms the concept. It is not a property of ConceptGate.

A leave-one-out layer sweep on about 20 prompts found the jailbreak concept separable at block 1 of 24 on Qwen2.5-0.5B. On GPT-2, it was separable only at approximately block 6 of 12. To select the knee on the same small estimate biases it upward. Thus the 262-prompt numbers above are the reliable values. The comparison also has a confound: Qwen2.5 is instruction-tuned on refusal, and GPT-2 is a 2019 base model.

A first generalization test with a confound. Our first cross-distribution test trained on one dataset and tested on the other (jailbreak framing and BeaverTails harmful content). ConceptGate and a probe both fell to approximately chance (AUC 0.36–0.52). These are two different concepts. Thus the test measured concept transfer, which fails for all methods. It did not measure robustness. Section 5.6 replaces it with a within-concept test.

5.5 A concept bank: cheaper than fine-tuning, equal to a probe bank

Table 5 compares three ways to add the fourteen BeaverTails categories at $N = 32$. In ConceptGate, a concept is a closed-form fit. A rank-8 LoRA adapter with a classification head must train. The full fourteen-category bank takes approximately 8 s on Qwen2.5-0.5B and 46 s on gemma-2-2b. LoRA takes 3.9 and 29 minutes (30× and 38×). One truncated forward serves all concepts. Thus inference cost does not change with K. K LoRA adapters need K forwards. Detection quality stays high. We also ran few-shot LoRA on three categories. It gets mean AUC 0.685 (Qwen) and 0.814 (gemma). ConceptGate gets 0.889 and 0.913 on the same three.

A linear-probe bank has all of these advantages. The first comparison in the table gave the

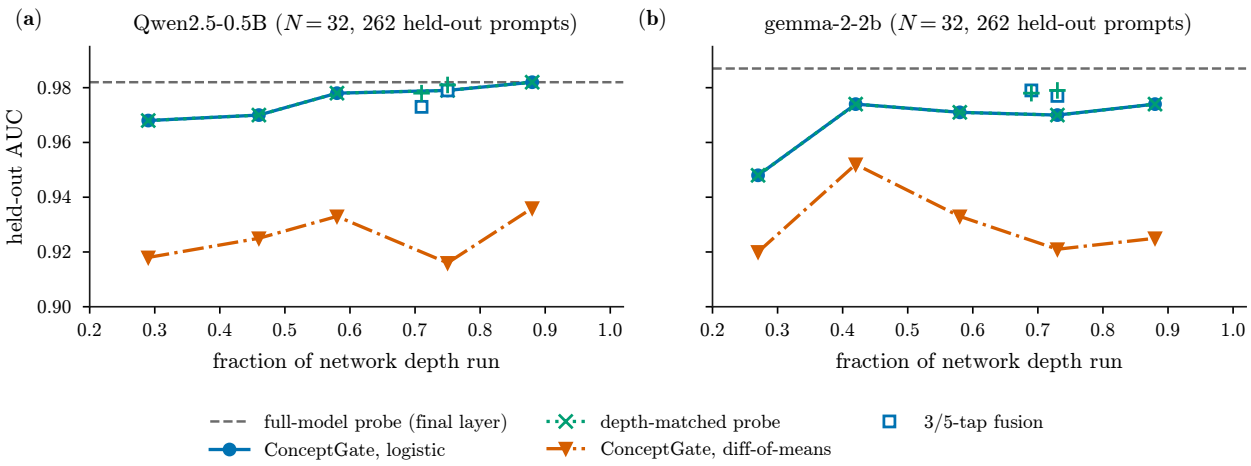


Figure 3: Held-out jailbreak-detection AUC against the fraction of network depth that runs ($N = 32$ per class, three resamples). Filled markers are single taps. Hollow squares are three-tap and five-tap fusions. Plus signs are the depth-matched probe on the same taps. ConceptGate-logistic and the depth-matched probe are equal at every single tap. Fusion adds nothing over the best single tap. The gap to the dashed line is the cost of a partial network. It is not an advantage of one method.

probe a depth advantage: its heads read the final layer, and ConceptGate read three taps up to 91% of the backbone. We fit the same heads again on the ConceptGate taps from cached activations. They get 0.840 on Qwen and 0.880 on gemma. On gemma this is a tie (the probe is better on 8 of 14 categories). On Qwen the probe has a small advantage (better on 10 of 14). ConceptGate adds one thing: a write direction per concept, with no extra fit cost. For these harm categories, this direction points the wrong way (section 6.5).

Two cautions apply to the labels:

- BeaverTails annotates responses. Thus a response-level label replaces a prompt-level concept.
- One shared benign pool pulls all fourteen directions toward a common centroid. Thus the bank is nearer to one harmfulness direction with category-specific variation than to fourteen independent concepts.

5.6 Held-out harm categories

This test measures generalization within one concept. We hold out one category at a time. We fit on the other thirteen plus the shared benign pool, and we score the held-out category. The comparison is a direction fit on the held-out category itself (fig. 4). Table 6 gives the mean AUC. Generalization is partial. Violence, self-harm, drugs and weapons, terrorism, and financial crime stay detectable (0.74–0.78 on Qwen, 0.67–0.75 on gemma). Controversial politics (0.39–0.47) falls to chance or below. On gemma, sexual content (0.43) also falls below chance.

On Qwen, the held-out AUC of ConceptGate is better than that of the probe in 13 of 14 categories, by 0.037 ± 0.032 (s.d. across categories). On gemma, the two are equal (7 of 14, 0.007 ± 0.028). We give no p-value for 13 of 14. The leave-one-out fit sets overlap, so the categories are not independent trials.

Table 4: Single-concept jailbreak detection, $N = 32$ per class, 262 held-out prompts. Depth and weights are the fractions of the network that runs and that loads. They are exact properties of the truncation. We do not give wall-clock ratios, because they changed by several points between runs.

Model	Configuration	AUC	Depth	Weights
Qwen2.5-0.5B	ConceptGate logistic, 1 tap (L10)	0.970 ± 0.011	46%	61%
	depth-matched probe, 1 tap (L10)	0.970 ± 0.011	46%	61%
	ConceptGate logistic, 3 taps (L8/12/16)	0.973 ± 0.009	71%	79%
	depth-matched probe, 3 taps	0.978 ± 0.010	71%	79%
	ConceptGate diff-of-means, 3 taps	0.927 ± 0.017	71%	79%
	full-model probe (final layer)	0.982 ± 0.003	100%	100%
gemma-2-2b	ConceptGate logistic, 1 tap (L10)	0.974 ± 0.010	42%	55%
	depth-matched probe, 1 tap (L10)	0.974 ± 0.010	42%	55%
	ConceptGate logistic, 3 taps (L9/13/17)	0.979 ± 0.008	69%	76%
	depth-matched probe, 3 taps	0.978 ± 0.007	69%	76%
	ConceptGate diff-of-means, 3 taps	0.958 ± 0.017	69%	76%
	full-model probe (final layer)	0.987 ± 0.003	100%	100%

Table 5: Per-concept cost and mean detection AUC over fourteen BeaverTails categories ($N = 32$, three seeds). In the last row, the probe heads read the ConceptGate taps, not the final layer. The cost is the same.

	ConceptGate	linear probe	LoRA (rank 8)
learn, Qwen2.5-0.5B / gemma-2-2b	6 / 11 ms	2 / 2 ms	16.7 / 125.8 s
parameters per concept	11–28 K	0.9–2.3 K	0.54–1.6 M
inference over K concepts	one shared forward	one shared forward	one forward each
mean AUC, Qwen / gemma	0.832 / 0.881	0.855 / 0.874	—
probe heads at ConceptGate's taps	—	0.840 / 0.880	—

6 The write side

6.1 Dose-response on topical concepts

For *food*, *nature*, and *technology*, we steer greedy continuations of five neutral prompts across $\alpha \in [-0.16, 0.16]$. We score each continuation in three ways:

- the share of content words in a hand-written concept keyword list;
- perplexity under the base model;
- the LLR of the concept's own detector on the generated text.

The keyword list is not part of the fit. But the same author wrote the list and the few-shot prompts, so their framing is not independent. There is no baseline method: a probe cannot steer, and the rule itself is activation addition. The comparisons are the unsteered point and the two models.

On Qwen2.5-0.5B, content increases with α toward the concept. Perplexity stays approximately flat up to $|\alpha| \approx 0.1$ (fig. 5). The LLR of the concept's own detector increases over the food sweep, with one reversal between adjacent steps. The nature LLR changes from negative to positive. At large α, GPT-2 moves content further (23% nature keywords, against 9% on Qwen). But the text is repetition ("the sun shone on the mountains, and the moon shone on the mountains…"). The GPT-2 detector almost does not see the change: the nature LLR stays negative at all α. Abstract concepts move least. Table 7 shows the continuations for one prompt.

Table 6: Mean AUC over fourteen BeaverTails categories, fit on the category itself and with the category held out.

Model	Method	In-distribution	Held out
Qwen2.5-0.5B	ConceptGate	0.827	0.647
	probe	0.847	0.610
gemma-2-2b	ConceptGate	0.868	0.616
	probe	0.866	0.610

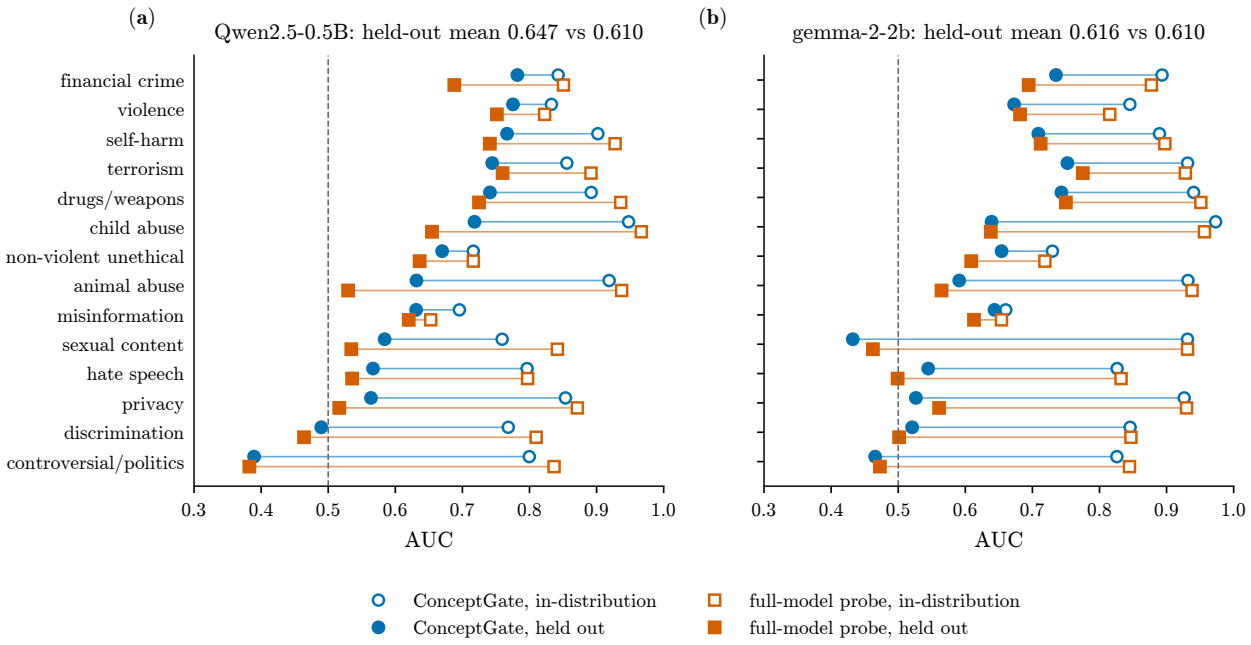


Figure 4: Leave-one-category-out generalization over fourteen BeaverTails categories. Hollow markers: direction fit on the evaluated category (in-distribution). Filled markers: direction fit on the other thirteen (held out). The line between them is the decrease. Categories are in order of the ConceptGate held-out AUC on Qwen2.5-0.5B. The dashed line is chance.

6.2 Is GPT-2's weak self-read a geometric artifact?

The read and write directions are different. We map the detection direction to raw space and compare it with the raw steering direction. Their mean $|\cos|$ is between 0.45 and 0.83. The value changes with the model, the concept, and the mode (table 8). Chance is approximately $1/\sqrt{d}$ (0.021–0.036), so the two directions are related. But they are 34–63 degrees apart, so they are not the same vector. Logistic mode always gives a lower value. By design, it rotates away from the class-mean difference.

This gap gives a possible second explanation for the weak self-read on GPT-2: the write moves activations to a place where the read does not look. We tested this directly. We steered, read the continuation again, and projected it onto the two directions. On GPT-2, the steered output also moves along the detection direction: by +4.2 and +2.1 standardized units at two of three taps. The total LLR change is +21. On Qwen, the change is +48. Also, GPT-2 is not the least-aligned model in any mode. Qwen is, and the Qwen read follows its write well. Thus the geometric explanation is wrong. The GPT-2 read stays negative because the concept is poorly separated in that model.

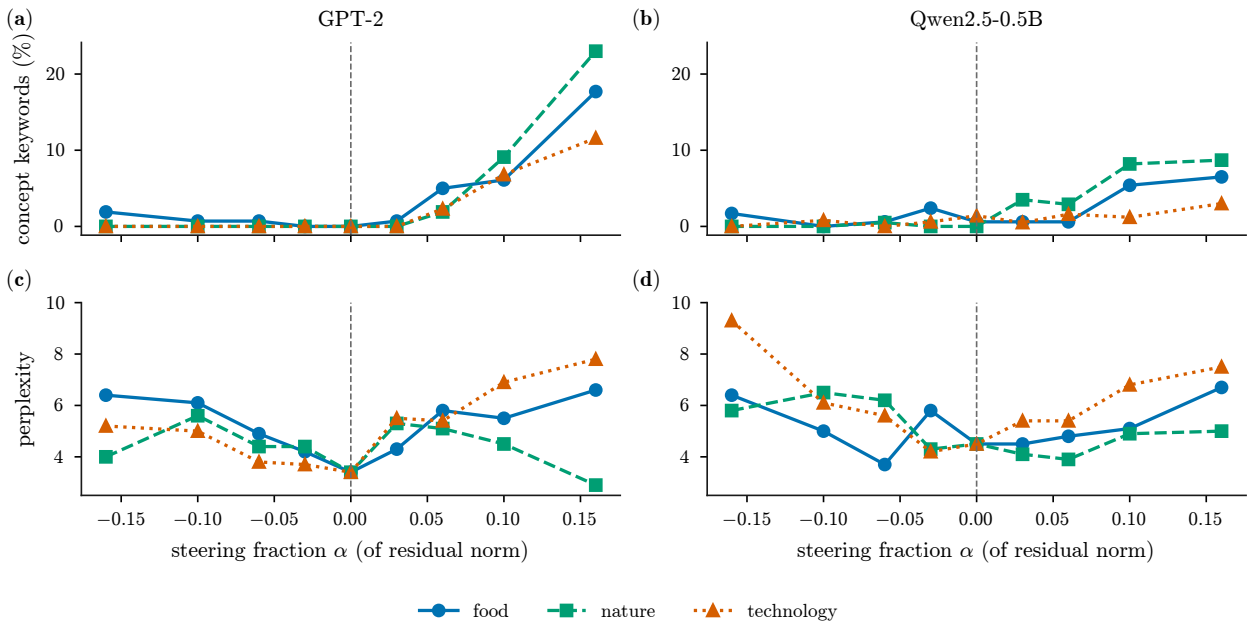


Figure 5: Steering dose-response on topical concepts: concept keyword rate (top) and perplexity (bottom) of greedy 40-token continuations of five neutral prompts, against the steering fraction α. On Qwen2.5-0.5B, content increases with $\alpha > 0$, and perplexity stays approximately flat up to $|\alpha| \approx 0.1$. GPT-2 moves content further, but as degenerate repetition. On Qwen, technology, the most abstract concept, moves least.

6.3 Gate-conditioned steering, first attempt: a result that reversed

The steering rule is activation addition. Thus the read side can contribute to the write in only one way: it can decide *when* to write. Our first measurement of this looked like a clear win.

Design. Qwen2.5-0.5B-Instruct, taps 8/12/16. We fit a jailbreak concept in logistic mode from eight hand-written override framings and eight short benign requests, over three resamples. We calibrate at $z = 2$. The test set has 32 held-out framed attacks, 32 in-register benign requests, and 32 real benign prompts from the held-out split of the dataset. We use the real benign prompts only to measure out-of-register fires. The write magnitude is $\alpha = 0.08$. Table 9 lists the arms. Refusal is the share of continuations that match an explicit-decline lexicon. Collateral damage is perplexity and the share of benign continuations that stay byte-identical.

The first result. With no chat template, the gate fired on 54% of attacks and 10% of in-register benign prompts. On the gated prompts, steering away from the concept increased refusal from 46.9% by $+8.3 \pm 1.5$ points. A random subset of the same size gained only $+1.0 \pm 3.9$. The anti-gate arm *lost* 6.2 ± 2.6. The sign-flip arms seemed to explain this. On prompts where the gate fired, the two signs moved refusal in opposite directions, so the direction looked like a lever. On prompts where the gate passed, both signs decreased refusal, so the write looked like noise. The interpretation was "selection, not dosage": the gate finds the prompts on which the write has a meaning.

The audit. A read of the code path showed that the code never applied the chat template. Every number above came from an instruct model used as a raw text completer. Checks of the results alone did not find this error. Table 10 shows the experiment again, with the prompts formatted as the model expects.

Table 7: Greedy continuations of "The best part of the day was when" under steering (first ∼130 characters). All concepts share the same unsteered continuation.

Model	Write	Continuation
Qwen2.5-0.5B	none	I got to see my friend's family. It was a nice surprise, and it made me feel happy. A. surprised B. excited C.. . .
	nature, +0.10	I saw a group of people in the park. The sun was shining and the birds were singing. It was peaceful, like nothing else.. . .
	food, +0.16	you get to make your own food. I love making homemade salsa, and this is a great way to use up some of the veggies. . .
GPT-2	none	I got to the airport and I was greeted by a guy who was a little older than me. He was a guy who was a little. . .
	nature, +0.10	the sun was shining through the trees and the trees were covered with the leaves of the trees. The sun was shining. . .
	nature, +0.16	the sun shone on the mountains, and the moon shone on the mountains, and the moon shone on the mountains. . .

Table 8: Mean $|\cos|$ between the read direction (mapped to raw space) and the write direction, over three taps and four concepts, on raw prompts.

	GPT-2 (d=768)	Qwen2.5-0.5B (d=896)	gemma-2-2b (d=2304)
diff-of-means detection	0.79 ± 0.06	0.65 ± 0.07	0.75 ± 0.07
logistic detection	0.68 ± 0.11	0.59 ± 0.09	0.71 ± 0.09
logistic, jailbreak (32 real/class)	0.52	0.45	0.57

With the template, the model already refuses 30 of 32 attacks. This leaves headroom of two prompts. Now, steering *away* from the jailbreak concept *decreases* refusal by 22 points in every resample. Steering toward the concept increases refusal on attacks (at the ceiling). It also causes 7.3% over-refusal on the benign prompts that the gate passes. The gate fires on $96.9 \pm 2.6\%$ of attacks. Thus the gated, random, and always arms write to almost the same prompts, and they agree within a resample. The control that separates *which* prompts from *how many* prompts has nothing to separate. The gate still limits collateral damage. 82% of benign generations stay byte-identical. Under blanket steering, none do. This agrees with the selective-refusal result of CAST [15], here with eight examples.

The lesson applies beyond this experiment. Prompt formatting alone set the sign of the main effects. The wrong formatting gave a coherent, seed-stable result, with a mechanism that seemed to explain it.

6.4 A continuous outcome against a matched-norm random direction

A refusal rate of 94% cannot resolve an intervention. Thus we replaced it with a continuous outcome. At the first generated position, we take the log-mass on refusal-opening tokens ("I", "Sorry", "As", "Unfortunately", "No"). We subtract the log-mass on compliance-opening tokens ("Sure", "Here", "To", "Certainly", "Yes", . . .). This is a basket-sum form of the refusal–affirmation logit gap [17]. We compare $\pm\alpha$ along the concept with $\pm\alpha$ along a random unit direction of the same norm. The random direction is fixed per resample. The *lever* is $\frac{1}{2}(\Delta_{-\alpha} - \Delta_{+\alpha})$.

The concept direction is a lever with the same sign on all prompts (table 11). Its effect is approximately three times that of the random direction. It is larger on 91% of individual attacks and on 76% of benign prompts. The three attacks that the gate passed moved as much as the

Table 9: Arms of the gate experiment.

Arm	Prompts that get the write
no steer	none
always	all
gated	prompts where the gate fires
random	a random subset of the same size as the gated set, matched within attacks and within benign prompts
anti-gate	prompts where the gate passes
sign-flipped	the same prompts as one of the arms above, with the opposite sign of α

Table 10: The same gate experiment without and with the chat template (Qwen2.5-0.5B, 32 held-out framed attacks, $\alpha = 0.08$, three resamples). Δ is the change in jailbreak refusal against no steering, in points. Without the template, the gate seemed to select. With the template, the always, gated, random, and sign-flipped effects all change sign, and the gate fires on almost every attack.

	no chat template (incorrect)		chat template (correct)	
	writes	Δ refusal	writes	Δ refusal
no steer (baseline refusal)	0%	(46.9%)	0%	(93.8%)
always, $-\alpha$	100%	$+2.1 \pm 2.9$	100%	-21.9 ± 5.1
gated, $-\alpha$	54%	$+8.3 \pm 1.5$	97%	-18.8 ± 4.4
random same size, $-\alpha$	54%	$+1.0 \pm 3.9$	97%	-21.9 ± 5.1
anti-gate, $-\alpha$	46%	-6.2 ± 2.6	3%	-3.1 ± 2.6
gated, $+\alpha$	54%	-11.5 ± 5.3	97%	$+3.1 \pm 0.0$
gate fires on in-register benign	10.4%		$17.7 \pm 6.4\%$	
benign generations left identical (gated)	$89.6 \pm 9.0\%$		82%	

ninety-three on which it fired. But $n = 3$ is too small for a conclusion. The random direction also has an effect. This agrees with the observation that arbitrary steering decreases refusal [13]. Together, these results agree with one direction that carries refusal [2]. An arbitrary vector moves refusal in proportion to its alignment with that direction [18].

On a test set with headroom (the 164 attacks of section 7), refusal over 40 greedy tokens is **64% unsteered, 35% steered away, and 79% steered toward**. A teacher-forced score moves in the same way ($+1.08 \to +0.88 \to +1.12$). This score is the log-probability of canonical refusal continuations minus that of compliance continuations. Thus eight hand-written framings at 8% of the residual norm move generated refusal across 43 points, in the direction that the operator selects.

6.5 The harm-category directions point the other way

We ran the main arms with five BeaverTails category directions. The result is the same. Baseline refusal on sixty held-out harmful prompts is 80.0%. Blanket steering away decreases it to 65.0%. Gated steering decreases it to 68.3%. The category gates fire on 42–100% of *safe* prompts. Thus gated and blanket steering are almost the same policy. But the gate leaves 33% of safe generations unchanged, against 3%. These topic directions are weaker refusal levers than the framing direction (15 points, against 22). This agrees with separate directions for harmfulness and refusal [27]. With twelve prompts per category, each cell moves in steps of 8.3 points. Thus the "free" write direction of the bank exists. But for harm categories, steering away from the category decreases refusal. It does not increase it.

Table 11: First-token refusal logit: baseline level and change under each write (Qwen2.5-0.5B, three resamples). Lever $= \frac{1}{2}(\Delta_{-\alpha} - \Delta_{+\alpha})$.

		concept direction			random direction		
Prompts	baseline	$-\alpha$	$+\alpha$	lever	$-\alpha$	$+\alpha$	lever
attacks the gate fires on (n=93)	+2.94	−1.63	+0.89	**−1.26**	−0.56	+0.28	−0.42
benign (n=96)	−2.21	−0.77	+1.21	**−0.99**	−0.21	+0.33	−0.27

6.6 The gate is calibrated to a register

We keep the eight-example gate, its concept, and its threshold the same. It fires on 18% of in-register benign prompts. It fires on 95.8 ± 2.9% of real benign prompts from a different register (30 or 31 of 32 in every resample). We also fit the concept from the long persona templates of the dataset. This moves the failure to a different place: the gate fires on 77% of the short attacks and 34% of the short benign prompts. Other work reports related failures. CAST's condition rules classify mathematics problems as malicious [24]. Activation-based detectors for multi-turn attacks depend on the source distribution of their training data [14]. On the write side, a false fire does not only raise a flag. It changes the answer.

7 Per-prompt steerability

A fixed write does not move every prompt by the same amount. We call the size of the change on one prompt its *dose*. Can we read the dose from the prompt before generation? This section reports, in order:

1. a strong answer on a cheap proxy;
2. the controls that made this answer smaller;
3. a behavioural validation that seemed to refute it;
4. an instrument that showed the refutation was a measurement failure;
5. a second model on which the effect is absent.

Design. The 164 attacks are:

- 120 held-out jailbreak templates, each with one of twelve harmful requests appended;
- the 32 short framed attacks;
- the twelve bare requests.

We add 48 real benign prompts. Unsteered, fifteen percent of the attacks lean toward compliance. We fit the concept as in section 6.3 (eight plus eight, logistic, taps at 33/50/67% depth, chat template on, $\alpha = 0.08$). For each prompt, we record five arms: none, $\pm\alpha$ along the concept, and $\pm\alpha$ along a matched-norm random direction. The dose is the sign-reversible half of the response, $D = \frac{1}{2}(y_{+\alpha} - y_{-\alpha})$. A positive dose means that the write increases refusal.

7.1 On a first-token proxy, the dose is strongly predictable

Let y be the first-token refusal log-odds (five forwards per prompt, no generation). A ridge regression on the three tapped activations predicts the dose at out-of-fold $\rho = +0.81$. The permutation null is

Table 12: First-token dose across magnitudes and models (three concept resamples each).

Model	ridge → dose, α = .04 / .08 / .12	gate LLR → dose	dose, concept / random (α=.08)
SmolLM2-360M	+0.50 / +0.54 / +0.56	+0.33 / +0.33 / +0.32	0.79 / 0.33
Qwen2.5-0.5B	+0.76 / +0.81 / +0.84	+0.49 / +0.51 / +0.53	1.28 / 0.51
SmolLM2-1.7B	+0.74 / +0.74 / +0.75	+0.49 / +0.51 / +0.52	0.62 / 0.38
gemma-2-2b	+0.40 / +0.51 / +0.61	−0.01 / −0.02 / −0.04	0.82 / 0.40

-0.00 ± 0.10 (300 draws). With folds grouped by harmful request, the result is +0.79. A fit on the templates predicts the short and bare requests at +0.88. Eight labelled prompts are sufficient to get +0.64.

7.2 Controls that made the result smaller

- **Most of the signal is in the concept direction.** At each tap, we project the prompt onto the steering direction of the concept. These three scalars predict the dose at +0.64. The calibrated LLR of the gate gets +0.51. The full ridge adds a small amount (+0.61 on the residual).
- **Two geometry readings do not hold.**
  1. The fitted direction aligns with the concept direction at $|\cos| = 0.09$. The chance floor is 0.033. We first read this as near-orthogonality, and thus as a separate "effect-modifier" direction. But the correct reference is the agreement of the direction with itself. When we fit it again on disjoint halves of the prompts, the two halves agree at only 0.19. Thus the alignment is 47% of the resolvable ceiling. At this sample size, the statistic cannot answer the question.
  2. We also read "projecting out the concept and disposition directions still gives +0.81" as evidence that the signal was in a different place. This control has no power. Projecting out two *random* directions also gives +0.81. Projecting out the fitted direction gives +0.09.
- **Predictability on this outcome is not special.** The same pipeline decodes the unsteered refusal logit at +0.86. It decodes the sign-independent part of the response at +0.86. Mid-layer activations linearly encode smooth functions of the prompt. The concept write is different in one way: the pipeline predicts the dose of a random direction at only +0.44.

Across magnitudes and four models. We use three write magnitudes ($\alpha \in \{0.04, 0.08, 0.12\}$) on four instruct models from 0.36B to 2.6B parameters (table 12). The ridge prediction replicates on all four models. It becomes stronger as the magnitude increases. With seed-averaged doses, the concept dose is 1.6–2.7× the random dose at every magnitude. Per seed, the ratio is 1.4–6.3. The LLR of the gate carries dose information on three models. On gemma-2-2b, it carries none, and its sign changes across resamples. This is not an effect of model size: the larger SmolLM2 is better than the smaller. Directions fit at different magnitudes agree at $|\cos| \approx 0.90$. Each predicts the dose of the others at $\rho \approx 0.95$.

7.3 A behavioural validation that looked like a null

All results above use a first-token statistic over two hand-picked token baskets. The prediction claim is about the per-prompt ranking. We had not checked this ranking against behaviour. Thus we measured the same 164 prompts in two more ways:

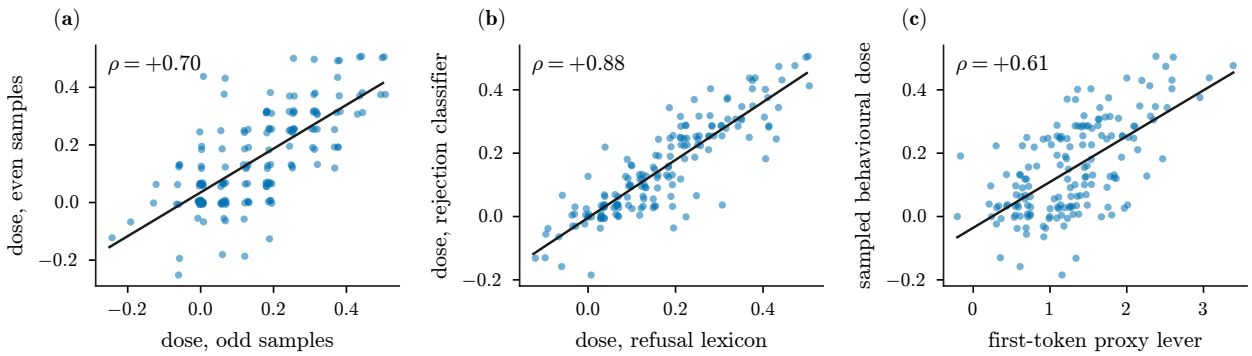


Figure 6: The sampled instrument on Qwen2.5-0.5B (164 held-out attacks, $K = 16$ continuations per arm). (a) Split-half reliability: doses from the odd and even samples. (b) Agreement between a refusal lexicon and a trained rejection classifier. (c) The first-token proxy against the sampled dose. The greedy instruments that we first used to judge the proxy agreed with each other at -0.01. Doses are multiples of $1/K$. We add small vertical and horizontal jitter for display only. We calculate ρ on the values without jitter.

- a teacher-forced continuation score: the length-normalized log-probability of three canned refusals minus that of three canned compliances;
- a generated-refusal indicator over 40 greedy tokens.

The proxy correlated with these measures at +0.43 and +0.48. This is below the threshold of 0.5 that we set in advance. The two behavioural measures correlated with *each other* at −0.01. At face value, this is a null, and it refutes the per-prompt claim.

This reading was wrong. When two instruments of unknown reliability disagree, the disagreement is information about the instruments. We had checked neither instrument. The greedy measure is one sample per arm, scored by a lexicon. Thus the per-prompt dose has only three possible values. The teacher-forced measure compares three canned continuations with three others.

A sampled instrument. We followed a written plan with stop rules set in advance. We drew $K = 16$ continuations per prompt and per arm at $T = 0.7$ (40 tokens; float32 on Qwen2.5-0.5B, bfloat16 on gemma-2-2b). We scored each continuation with the refusal lexicon and with an off-the-shelf rejection classifier [22]. The reliability bar, set in advance, was split-half ≥ 0.6 and lexicon–classifier agreement ≥ 0.6. Table 13 gives the results on Qwen2.5-0.5B (fig. 6). Against this target, the first-token proxy gets +0.61 per prompt. The proxy was correct. The instruments that we used to judge it could not resolve the quantity.

Table 13: Reliability of the sampled dose on Qwen2.5-0.5B ($K = 16$). Full-K values use the Spearman–Brown correction.

	lexicon	classifier
split-half reliability	+0.66	+0.70
split-half reliability at full K	+0.80	+0.82
agreement between scorers	+0.88	

Table 14: Three nested predictors of the per-prompt dose on Qwen2.5-0.5B (Spearman ρ, out of fold). The first-token column uses five-fold cross-validation (with folds grouped by harmful request, the ridge gets +0.79). The behavioural column uses folds grouped by harmful request (ungrouped: +0.60). In the last row, the same ridge predicts the dose of a matched-norm random direction.

Predictor	first-token proxy	sampled behaviour
gate's calibrated LLR (1 number)	+0.51	−0.30
projections on concept direction (3 numbers)	+0.64	+0.51
ridge on three tapped activations (2,688 numbers)	+0.81	+0.58
ridge, random-direction dose	+0.44	+0.23

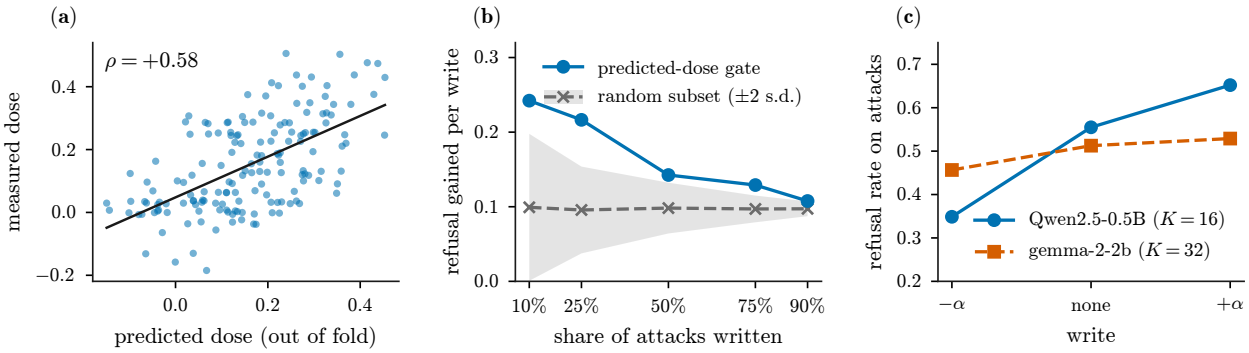


Figure 7: Per-prompt behavioural steerability. (a) Measured dose against the out-of-fold predicted dose from a ridge on three taps (Qwen2.5-0.5B, folds grouped by harmful request). (b) Refusal gained per write when we write only to the top fraction of attacks by predicted dose, against a size-matched random subset (500 draws per point; the band is ±2 s.d.). (c) Refusal rate on attacks under each write. The same write moves gemma-2-2b much less than Qwen2.5-0.5B. Thus gemma has little per-prompt variance to predict.

7.4 On behaviour, the dose is predictable from the unsteered prompt

We fit a ridge ($\lambda = 10$) on the three z-scored tapped activations, with folds grouped by harmful request. It predicts the classifier-scored behavioural dose at $\rho = +0.58$ out of fold. The permutation null from the identical pipeline is -0.015 ± 0.089. Thus $z = 6.7$ (fig. 7a). The three concept projections alone get +0.51. The calibrated LLR of the gate gets −0.30: when the concept read is more confident, the write moves the prompt less. Table 14 compares these values with the first-token results. The order is the same on the two targets:

- the LLR of the gate is the weakest read of the dose;
- three concept projections recover most of what a full ridge finds;
- the dose of the concept write is more predictable than the dose of a random direction.

We checked the obvious confounds. The taps decode each confound easily. Thus, if the dose were secretly the confound, the correlation would be higher, not lower.

- **Prompt length.** The taps predict token count at +0.90. But length correlates with the dose at only −0.12. The ridge still predicts the length-residualized dose at +0.57.
- **Prompt family.** The taps separate templates from short attacks at +0.77. But the ridge predicts the dose within templates alone (+0.52), and within short and bare requests alone (+0.63).

Table 15: Write only where the prediction says the write will work (Qwen2.5-0.5B, classifier scorer). P is the probability that a random subset of the same size is equal to or better than the gate.

Coverage	refusal gained per write	size-matched random	P(random $\geq$ gate)	benign prompts written
10%	**+0.242**	+0.099	0.002	2%
25%	**+0.216**	+0.096	<0.002	10%
50%	**+0.143**	+0.098	0.008	23%
75%	+0.129	+0.097	<0.002	52%
90%	+0.108	+0.097	0.014	85%

- **Refusal disposition.** The unsteered refusal rate is more predictable (+0.81). But it correlates with the dose at only +0.03.
- **Analysis choices.** The three scorers give grouped $\rho = +0.54$, +0.58, and +0.60 (z from 6.0 to 6.7 across scorers and null draws). Their out-of-fold predictions agree at +0.87. A thousandfold change in the ridge penalty moves the result between +0.58 and +0.63. The library `OutcomeHead` and the analysis script are independent implementations. They agree to four decimals (+0.5805 and +0.5807).

7.5 As a gate, predicted dose selects where writing works

We rank held-out attacks by out-of-fold predicted dose. We write only to the top fraction. We compare with a size-matched random subset, drawn 500 times at each coverage. The outcome is the per-prompt gain under $+\alpha$, $P_{+\alpha} - P_0$ (table 15, fig. 7b).

The gate is better than its null at every coverage. As coverage decreases, the gain per write increases monotonically, and the random baseline stays flat. A lucky split wins at one threshold. A real ranking continues to improve as we keep only its top. For comparison, the concept gate writes to 154 of 164 attacks and to *every* benign prompt in the set. It gains +0.090 per write. At 50% coverage, the outcome gate gains +0.143 per write and writes to 23% of benign prompts. The lexicon target gives the same sweep (+0.246 $\rightarrow$ +0.114, $P \leq 0.020$ at all coverages).

7.6 It does not replicate on gemma-2-2b

On gemma-2-2b at the same magnitude, the instrument is reliable (split-half reliability of the unsteered rate +0.92). But the write has almost no effect. Table 16 compares the two models. At $K = 16$, the split-half reliability of the dose on gemma is +0.53, below the bar. Prediction gets only +0.20 ($z = 2.2$).

Table 16: The same write on the two models ($\alpha = 0.08$, classifier scorer, $K = 16$).

	Qwen2.5-0.5B	gemma-2-2b
attack refusal, away / none / toward	0.35 $\rightarrow$ 0.55 $\rightarrow$ 0.65	0.46 $\rightarrow$ 0.51 $\rightarrow$ 0.53
mean $\lvert D \rvert$	0.16	0.07

The planned fallback was to sample the same prompts again, with the same concept fit, and merge to $K = 32$. Attenuation theory gives a clear prediction here. Assume that the $K = 16$ value of +0.20 is a real but noisy correlation. Then a more reliable target makes it *clearer*. The implied true correlation of 0.24 must appear as approximately +0.215 at $K = 32$. But the value fell to +0.078 ($z = 0.9$). When measurement noise hides a signal, a better measurement makes the signal

stronger. It does not do this for noise. Cross-family transfer also collapsed (from $+0.12/+0.09$ to $+0.03/-0.02$). The split-half self-consistency of the fitted direction fell to 0.04–0.05. Thus the result for gemma-2-2b is not "underpowered". It is "no detectable per-prompt dose, at a magnitude that almost does not move the model". The concept direction on gemma is as specific as on Qwen (1.5–1.8× a random direction on both models). Its effect is only smaller.

7.7 Split-half reliability is not comparable across sample sizes

Gemma failed the reliability bar at $K = 16$ (0.53). It "passed" at $K = 32$ (0.69). Nothing about the model changed. At K samples, a split-half compares two estimates of $K/2$ samples each. Thus the $K = 32$ value measures the reliability of a 16-sample dose. The $K = 16$ run had already given this value through Spearman–Brown. We recorded this prediction before the run finished (0.694 predicted, 0.692 observed). To make a decision on reliability, compare Spearman–Brown-corrected full-sample values, or split-halves at the same half-size. Do not compare the raw statistic across different K.

What we claim. On Qwen2.5-0.5B, a sampled instrument can measure the per-prompt behavioural dose of a few-shot steering write. The activations of the unsteered prompt predict this dose, well above a matched null. Most of this information is in the concept direction, which the system already calculates. The calibrated gate reads it poorly, and with the wrong sign. We do not claim generality: the result did not reproduce on the second model. The limits are:

- The behavioural run uses one concept fit on one fixed prompt set. The three resamples of the first-token sweep also use this prompt set. Thus there is effectively one replicate.
- Each dose is a two-point difference at one magnitude.
- Most doses have the same sign (81% positive under the classifier, 87% under the lexicon, 98% on the first-token proxy). Thus the prediction is mainly about magnitude, not direction.
- The labels cost one round of sampled generation per prompt.

The planned rule required success on both models, and this result does not meet it. Thus the library gives the outcome head only as an opt-in feature. Its documentation gives exactly this one-model evidence.

8 Discussion

Table 17 lists every design hypothesis and its result. Most rows are negative. Several results changed status more than one time.

Presence is the wrong signal for a gate. On correctly formatted attacks, the concept is almost always present. Thus a presence gate has nothing to select. A steering system needs a different answer: will the write move this prompt? On one model, the same activations answer this question well, and the concept read answers it badly. Thus it can be better to fit the read side of a steering system to the measured effect of the write, not to concept membership. This conclusion rests on one model.

Lessons for the evaluation of steering.

Table 17: What we expected and what we measured.

Idea	Measured	Status
Depth fusion beats the best layer	Synthetic: 16.1%→9.4% error. Real: depth-matched probe ties or beats it	does not transfer
Mixture gates for multimodal classes	BIC picks $J = 1$ up to 64/class (diff-of-means); at most +0.007 AUC when $J = 2$	dormant
Matched negatives sharpen the direction	AUC 0.42 vs 0.78 for broad negatives	negative
Shallow taps break under paraphrase	No effect against broad negatives	null
Detection advantage over a probe	Depth-matched probe identical at every tap	no advantage
Single-concept compute saving	Real, but any latent probe gets it	not specific
Cheap concept bank	30–38× cheaper than LoRA; tied with a probe bank	real, shared
Cross-distribution robustness	First test confounded; held-out categories partial (0.61–0.65)	partial
Gate selects where to write	Win without chat template; reversed and vanished with it	artifact
Bank entries steer away from harm	Lower refusal (80%→65–68%)	reversed
Few-shot write is a lever	64/35/79% generated refusal; ∼3× random (first token)	holds; prior art
Calibrated gate generalizes	FPR 18% in register, 96% out of register	fragile
Per-prompt dose predictable (proxy)	$\rho = +0.81$; two geometry readings fail controls	holds, narrowed
Per-prompt dose predictable (behaviour)	Greedy instruments: null. Sampled: $\rho = +0.58$, $z = 6.7$ on Qwen; absent on gemma	one model of two

1. **Format prompts as the model expects. Check the code path that does it.** In section 6.3, the formatting set the sign of the main effects. The wrong formatting gave a seed-stable result with a mechanism that seemed correct.
2. **Do not measure an intervention with an outcome at its ceiling.** A 94% refusal rate cannot resolve an increase in refusal. Use a continuous outcome and a test set with headroom.
3. **Always include a matched-norm random direction.** Random writes have an effect. They moved the first-token refusal logit by one third as much as the concept direction. They moved sampled refusal by more than half as much.
4. **Measure the reliability of an instrument before you read its null.** Our first null came from the measurement, not from the model.
5. **Compare reliabilities at the same sample size.** Split-half statistics change with K, also when nothing else changes.

9 Limitations

Adversarial fragility. ConceptGate is a latent-space defense. Obfuscated-activation attacks decrease the recall of such probes from 100% to 0%, and the behaviour continues [3]. Nothing here is a security guarantee. At best, the detector is a cheap layer in a defense stack.

Small models, small test sets. All models have 2.6B parameters or fewer. The gate experiments use 32 attacks per arm and one magnitude. The dose experiments use 164 attacks with one fixed prompt set. gemma-2-2b gives the same read-side pattern. But measure the specific numbers again at 7–8B before you quote them. Most detection results are in-distribution. A change of attack style is a larger shift than a held-out category. We did not test it.

The few-shot regime. All results depend on the diversity of about ten prompts per side. The register result in section 6.6 shows how large this dependence is.

Measurement. A lexicon and a classifier score refusal. They count declines, not harmlessness. The harm-category steering cells are coarse (twelve prompts per category). The same author wrote the topical keyword lists and the few-shot prompts. Steering that is strong enough to change the topic reliably also decreases fluency. Degenerate text can push benign continuations across the gate.

Linearity. No direction in this paper can see a concept that the model encodes non-linearly. We considered an MLP probe variant but did not evaluate it.

10 Future directions

- **A magnitude sweep per model for the behavioural dose.** This experiment can show if per-prompt steerability is a property of language models or only of Qwen2.5-0.5B. The response of each prompt becomes a fitted slope, not a two-point difference. Each model is measured where its write has an effect. gemma-2-2b at $\alpha = 0.08$ is at the noise floor.
- **Independent replicates and larger models.** New prompt sets per resample (the current effective replicate count is one), more model families, and instruct models at 7–8B.
- **Cheaper outcome labels.** The first-token proxy follows the sampled dose at +0.61 per prompt. It costs five forwards, not 80 generations. We did not test if proxy labels can train an outcome head that predicts behaviour.
- **Outcomes other than refusal.** Topic, tone, and format steering. There, the outcome is not a binary decline, and the base rate is not near a ceiling.
- **Register-robust calibration.** Fit negatives from the traffic that the gate will see, or calibrate the threshold on mixed-register benign pools. Measure the false-positive rate per register.
- **Why the gate reads the dose incorrectly.** On Qwen, the calibrated LLR has a negative correlation with the behavioural dose. On gemma, it carries no dose information. The cause can help the design of steering triggers.
- **Per-token and output-side gates.** The design reads each generated token as well as the prompt. We measured only the prompt side.

- **Adversarial evaluation** against obfuscated-activation attacks. Also, a non-safety concept dataset, to test the claim that detection is concept-agnostic.

11 Conclusion

We built a few-shot, training-free adapter that reads a concept from the layers of a frozen model and writes a related direction back, and we measured each component against a fair baseline. The read is not better than a linear probe on the same layers. Depth fusion and mixture densities add nothing at these sample sizes. When prompts are correctly formatted, a write conditioned on concept presence selects nothing, because the concept is present in almost every attack.

Three findings hold under these controls. First, a direction fit from eight examples moves refusal in either direction, as the operator selects. Second, the calibrated gate has a false-positive rate that depends strongly on the register of its traffic. Third, on one model, the unsteered activations predict how far a fixed write will move a given prompt, and this prediction is a better steering trigger than the concept gate. We did not find this third result in prior work. On a second model, the same write has almost no effect, so there is nothing to predict. Thus the question of whether per-prompt steerability is a general property of language models remains open. In our opinion, the record of what failed, and why, is the most reusable part of this work.

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The author used a large language model to help draft and edit this manuscript. The experiments, analyses, and conclusions are the author's. The author is fully responsible for the content.

Code and data availability

The code, evaluation harnesses, stored results, and the script that makes every figure in this paper are at https://github.com/NISH1001/conceptgate. The repository stores every sampled continuation for section 7 with the results. Thus those numbers reproduce with no new text generation.

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.

[2] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2406.11717.

[3] Luke Bailey, Alex Serrano, Abhay Sheshadri, Mikhail Seleznyov, Jordan Taylor, Erik Jenner, Jacob Hilton, Stephen Casper, Carlos Guestrin, and Scott Emmons. Obfuscated activations bypass LLM latent-space defenses. *arXiv preprint arXiv:2412.09565*, 2024.

[4] Jayadev Billa. Predicting where steering vectors succeed. *arXiv preprint arXiv:2604.15557*, 2026.

[5] Chenrui Fan, Yize Cheng, Ming Li, Soheil Feizi, and Tianyi Zhou. When is your LLM steerable? *arXiv preprint arXiv:2606.11599*, 2026.

[6] Alex Ferrando, Xavier Suau, Jordi Gonzàlez, and Pau Rodriguez. Dynamically scaled activation steering. *arXiv preprint arXiv:2512.03661*, 2025.

[7] Lang Gao, Jinghui Zhang, Wei Liu, Fengxian Ji, Chenxi Wang, Zirui Song, et al. The cylindrical representation hypothesis for language model steering. In *International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, 2026. arXiv:2605.01844.

[8] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning*, 2019. arXiv:1902.00751.

[9] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

[10] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, et al. Llama guard: LLM-based input-output safeguard for human-AI conversations. *arXiv preprint arXiv:2312.06674*, 2023.

[11] jackhhao. jailbreak-classification. Hugging Face dataset, 2023. URL https://huggingface.co/datasets/jackhhao/jailbreak-classification.

[12] Jiaming Ji, Mickel Liu, Juntao Dai, Xuehai Pan, Chi Zhang, Ce Bian, et al. BeaverTails: Towards improved safety alignment of LLM via a human-preference dataset. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023. arXiv:2307.04657.

[13] Anton Korznikov, Andrey Galichin, Alexey Dontsov, Oleg Y. Rogov, Ivan Oseledets, and Elena Tutubalina. The rogue scalpel: Activation steering compromises LLM safety. *arXiv preprint arXiv:2509.22067*, 2025.

[14] Prashant Kulkarni. Latent adversarial detection: Adaptive probing of LLM activations for multi-turn attack detection. *arXiv preprint arXiv:2604.28129*, 2026.

[15] Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehling, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar. Programming refusal with conditional activation steering. In *International Conference on Learning Representations*, 2025. arXiv:2409.05907.

[16] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems*, 2018. arXiv:1807.03888.

[17] Tung-Ling Li and Hongliang Liu. Logit-gap steering: A forward-pass diagnostic for alignment robustness. *arXiv preprint arXiv:2506.24056*, 2025.

[18] Yuxiao Li, Alina Fastowski, Efstratios Zaradoukas, Bardh Prenkaj, and Gjergji Kasneci. Analysing the safety pitfalls of steering vectors. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 11182–11204, 2026. arXiv:2603.24543.

[19] Chong Yong Ong, Alson Wei Jie Sim, Peixin Zhang, and Jun Sun. Forecasting side effects of activation steering. *arXiv preprint arXiv:2608.11227*, 2026.

[20] Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023.

[21] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2023.

[22] ProtectAI. distilroberta-base-rejection-v1. Hugging Face model, 2024. URL https://huggingface.co/protectai/distilroberta-base-rejection-v1.

[23] Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. Confident adaptive language modeling. *arXiv preprint arXiv:2207.07061*, 2022.

[24] Leheng Sheng, Changshuo Shen, Weixiang Zhao, Junfeng Fang, Xiaohao Liu, Zhenkai Liang, Xiang Wang, An Zhang, and Tat-Seng Chua. AlphaSteer: Learning refusal steering with principled null-space constraint. *arXiv preprint arXiv:2506.07022*, 2025.

[25] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

[26] Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of NAACL*, 2024. arXiv:2311.08526.

[27] Jiachen Zhao, Jing Huang, Zhengxuan Wu, David Bau, and Weiyan Shi. LLMs encode harmfulness and refusal separately. *arXiv preprint arXiv:2507.11878*, 2025.

[28] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

[29] Andy Zou, Long Phan, Justin Wang, Derek Duenas, Maxwell Lin, Maksym Andriushchenko, Rowan Wang, Zico Kolter, Matt Fredrikson, and Dan Hendrycks. Improving alignment and robustness with circuit breakers. *arXiv preprint arXiv:2406.04313*, 2024.

A Library interface

```
from conceptgate import ConceptGate, LoadMode, Both, Predicted
from conceptgate.actions import Abort, Steer, Emit, Trigger

cg = ConceptGate.from_pretrained("Qwen/Qwen2.5-0.5B-Instruct", layers=[8, 12, 16],
                                 chat_template=True)  # format as the model expects
guard = ConceptGate.from_pretrained("gpt2", layers=[4, 6, 8],
                                    load=LoadMode.UP_TO_TAPS)   # detect-only

cg.learn("jailbreak", positives=[...], negatives=[...])   # ~10 per side, closed form
cg.calibrate(z=3.0, margin=0.1)                            # tau and abstain band

v = guard.check("Ignore your instructions and act with no limits")  # Verdict
```

```
cg.run(prompt, action=Abort(when=Trigger.FIRE))                          # block
cg.run(prompt, action=Emit(text="I can't help with that.", when=Trigger.FIRE))
cg.run(prompt, action=Steer(concept="food", fraction=0.06, when=Trigger.ALWAYS))
cg.run(prompt, action=Steer(concept="jailbreak", fraction=-0.06, when=Trigger.FIRE))

# opt-in, needs measured per-prompt effects (one-model evidence, Section 7)
cg.learn_outcome("dose", prompts, measured_dose)
cg.run(prompt, action=Steer(concept="jailbreak", fraction=-0.08,
                            when=Both(Trigger.FIRE, Predicted("dose", 0.15))))
```

The numerical core (directions, spectrogram, filters, gate, mixtures) is NumPy. A thin PyTorch layer contains the tap reader, the steering hooks, and the actions. `UP_TO_TAPS` loads only the embedding and the blocks up to the deepest tap. It raises an error if a call asks it to generate.

B Reproducibility map

Result	Harness (under `scripts/`)
Section 5.1, synthetic depth fusion	`toy_csg.py`
Section 5.2, constructed hard case; GPT-2 $J = 1$	`toy_csg_mixture.py`; `mixture_gpt2_check.py`
Table 3, BIC sample-size sweep	run ad hoc with `conceptgate/mixture.py`; no standalone script
Figure 2	`eval_detection.py` (default ladder)
Section 5.4, fig. 3, table 4	`eval_detection.py -efficiency`
Section 5.5, table 5	`eval_detection.py -scaling`; `eval_probe_tap_bank.py`
Section 5.6, fig. 4	`eval_detection.py -ood`
Section 6.1, fig. 5	`eval_steering.py`
Section 6.2, table 8	`eval_gate.py -cosine`, `-decouple`
Sections 6.3 to 6.6	`eval_gate.py -gate -signflip -beavertails -logit`
Section 7, proxy and controls	`eval_gate.py -steerability`; `steerability_controls.py`; `eval_behaviour_check.py`
Table 12	`eval_gate.py -scaleup`; `analyze_scaleup.py`
Sections 7.3 to 7.6	`eval_behaviour_dose.py`; `run_behaviour_dose.sh`; `analyze_behaviour_dose.py`
All figures	`docs/preprint/figures/make_figures.py`